

AP CALCULUS AB - UNIT 1

1.9 Connecting Multiple Representations of Limits

Lesson Overview

- Extract one-sided limits, two-sided limits, and point values from any representation.
- Translate accurately among formula, graph, table, words, and notation.
- Distinguish exact conclusions from numerical estimates.
- Audit whether multiple representations are locally coherent.

Key Knowledge and Vocabulary

Representation. A graph, table, formula, verbal statement, or symbolic statement that encodes mathematical behavior.

Local information. Limit questions depend on behavior arbitrarily close to the target.

Point information. $f(c)$ is a separate datum and may disagree with the limit.

Coherence. Different representations are coherent when they encode the same one-sided limits and point value.

Precision. Algebra may establish an exact value; a graph or finite table generally provides an estimate.

Explanation and Strategy

Step 1. Create a fact ledger: left limit, right limit, two-sided conclusion, and point value.

Step 2. When translating a formula into a graph, preserve excluded inputs and special assigned values.

Step 3. For composite limits, include the direction from which the inner function reaches the outer target.

Solved Examples

Worked Example: Formula to graph facts

Let $f(x) = \frac{x^2-1}{x-1}$ for $x \neq 1$ and $f(1) = 6$. Describe the graph and state the limit.

Solution. For nearby $x \neq 1$, $f(x) = x + 1$, so the graph is the line $y = x + 1$ with an open circle at $(1, 2)$ and a filled point at $(1, 6)$. The limit is 2.

Worked Example: Table to notation

Use Representation Set R to write a limit statement at $x = 0$.

Solution. Values from both sides approach 2, so $\lim_{x \rightarrow 0} p(x) = 2$. The separate assigned value is $p(0) = 5$.

Worked Example: Directional composition

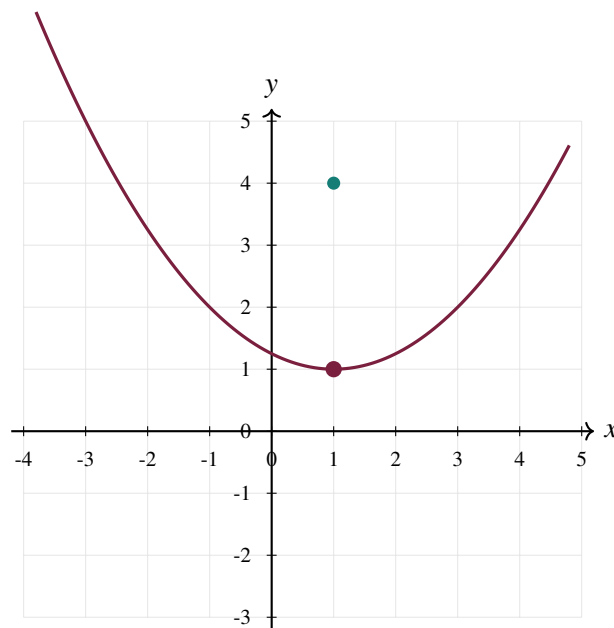
If $g(x) \rightarrow 1^-$ and a graph of F has $\lim_{t \rightarrow 1^-} F(t) = 4$ but $\lim_{t \rightarrow 1^+} F(t) = -2$, find $\lim F(g(x))$.

Solution. The inner values land at 1 from below, so the outer left-hand limit applies and the composite limit is 4.

Leveled Practice

Shared Representation / Data

Representation Set R



x	-0.10	-0.01	-0.001	0.001	0.01	0.10
$p(x)$	2.31	2.0301	2.003001	1.997001	1.9701	1.71

The formula $p(x) = 2 - 3x + x^2$ for $x \neq 0$ and the assigned value $p(0) = 5$ describe a separate function with the same local pattern as the table.

Easy Level

Foundational fluency. Focus on notation, definitions, and direct procedures.

1. A graph approaches 3 from both sides at $x = 2$ and has a filled point at 7. Write the limit and point value.

2. A table approaches -4 from both sides at $x = 1$. Translate this into notation.

3. The formula simplifies to $x + 5$ for $x \neq 3$. What local graph feature occurs at $x = 3$ if the original formula is undefined there?

4. Which representation most directly displays an exact algebraic cancellation?

Medium Level

Apply the idea in one or two connected steps.

5. Use Representation Set R to state the left-hand limit, right-hand limit, and two-sided limit at 0.

6. Use Representation Set R to state the assigned point value in the formula description.

7. Describe a table coherent with $\lim_{x \rightarrow 4^-} f(x) = 1$ and $\lim_{x \rightarrow 4^+} f(x) = 6$.

8. Translate $\lim_{x \rightarrow -2^+} g(x) = +\infty$ into a graphical description.

Hard Level

Connect representations, conditions, and justification.

9. A graph shows a hole at height 5, but a table rounded to the nearest integer shows 4 on the left and 6 on the right. Explain how both could describe the same limit.

10. A formula implies a removable hole at $x = 1$, but a supplied graph shows a jump. What should be concluded?

11. If $g(x) \rightarrow 2^+$ and F has a jump at 2 with right-hand limit 7, translate the composite limit into notation.

12. Why can a graph and table support the same estimated limit while reporting different last decimals?

Difficult Level

Use nonroutine structure and precise mathematical reasoning.

13. Construct a formula whose graph has a hole at $(2, 5)$ and a filled point at $(2, -1)$.

14. A table samples a function along only one sequence approaching c . Explain, using representation language, why this cannot establish coherence with a proposed graph.

15. If two formulas agree for all $0 < |x - c| < \delta$ but differ elsewhere, compare their limits at c .

Challenge Level

Synthesize, construct, generalize, or prove.

16. Create five coherent representations for a function with $\lim_{x \rightarrow 1} f(x) = 4$ and $f(1) = -3$: notation, words, formula, graph description, and table description.

17. Can two functions have identical graphs at every displayed pixel and different limits at the target? Explain.

AP-Style Practice

Multiple-Choice Practice – No Calculator

Choose the best answer. Use exact values unless a decimal approximation is requested.

1. Which pair of statements is coherent?

- (A) $\lim_{x \rightarrow 2} f(x) = 3$ and a graph approaching 5 from both sides
(B) $f(2) = 7$ and a filled point at $(2, 7)$
(C) $\lim_{x \rightarrow 2^-} f(x) = 1$ and a left table approaching 4
(D) A formula equal to $x + 1$ near 2 and a hole at height 8

2. A table approaches 6 from both sides while the formula is undefined at the target. Which is the best conclusion?

- (A) The limit must be undefined.
(B) The limit is estimated as 6 and the point value is undefined.
(C) The point value is 6.
(D) The limit is infinite.

3. If an inner function approaches a jump point from the right, the composite limit uses

- (A) the outer left-hand limit
(B) the outer right-hand limit
(C) the outer point value only
(D) the average of both sides

Multiple-Choice Practice – Calculator Required

Choose the best answer. Use exact values unless a decimal approximation is requested.

4. A graphing calculator displays a line through $x = 3$ for $f(x) = \frac{x^2 - 9}{x - 3}$. What information may be hidden?

- (A) A removable hole at $x = 3$
(B) A vertical asymptote at $x = 0$
(C) A jump at every integer
(D) A horizontal asymptote at $y = 0$

5. Calculator tables from both sides approach 1.41421. Which exact value is a plausible conjecture requiring algebraic confirmation?

- (A) $\sqrt{2}$
(B) π
(C) e
(D) $3/2$

Free-Response Practice – No Calculator

A function is represented by $f(x) = \frac{x^2 - 4}{x - 2}$ for $x \neq 2$ and $f(2) = -5$.

(a) State the limit and point value at $x = 2$. [2 points]

(b) Describe the corresponding graph near $x = 2$. [2 points]

(c) Give a four-row table description coherent with the formula. [2 points]

Free-Response Practice – Calculator Required

A calculator table for q near $x = 0$ gives values 2.0301, 2.003001, 1.997001, 1.9701 at inputs $-0.01, -0.001, 0.001, 0.01$.

(a) Estimate the two-sided limit. [1 points]

(b) Give one formula consistent with the data to the shown precision. [2 points]

(c) Explain why the table cannot determine $q(0)$. [2 points]